

Credit Risk Modeling Agentic AI : Salary Prediction Module

By

Ethel Ng Yi Yan



FACULTY OF COMPUTING AND
INFORMATION TECHNOLOGY

TUNKU ABDUL RAHMAN UNIVERSITY OF
MANAGEMENT AND TECHNOLOGY
KUALA LUMPUR

ACADEMIC YEAR
2025/26

Credit Risk Modeling : Salary Prediction Module

By

Ethel Ng Yi Yan

Supervisor: Assoc. Prof. Ts. Dr Yu Yong Poh

A project report submitted to the
Faculty of Computing and Information Technology
in partial fulfillment of the requirement for the
Choose an item.

Faculty of Computing and Information Technology
Tunku Abdul Rahman University of Management and Technology
Kuala Lumpur

Copyright by Tunku Abdul Rahman University of Management and Technology.

All rights reserved. No part of this project documentation may be reproduced, stored in a retrieval system or transmitted in any form or by any means without prior permission of Tunku Abdul Rahman University of Management and Technology.

Declaration

The project submitted herewith is a result of my own efforts in totality and in every aspect of the project works. All information that has been obtained from other sources had been fully acknowledged. I understand that any plagiarism, cheating or collusion or any sorts constitutes a breach of TAR University rules and regulations and would be subjected to disciplinary actions.

Ethel

Ethel Ng Yi Yan

ID: 24WMR08861

Table of Contents

1 Introduction	2
1.1 Project Overview.	2
1.2 Project Background.	2
1.2.1 Problem Statement	3
1.3 Project Objective	5
1.4 Scope of Project	5
1.5 Advantages and Contribution	7
1.5.1 Alignment with Sustainable Development Goals (SDGs) and Malaysia Madani	8
1.6 Project Plan & Gantt Chart	9
1.6.1 Project Plan	9
1.6.2 Gantt Chart	11
1.7 Project Team and Organisation	11
1.7.1 Organisation Chart	11
1.8 Chapter Summary and Evaluation	12
2 Literature Review	15
2.1 Introduction	15
2.2 The landscape of Credit Assessment	16
2.2.1 Traditional Credit Scoring and Its Limitations	16
2.2.2 Transition to Data-Driven and AI-Based Models	17
2.3 Salary Prediction / Income Forecasting	17
2.3.1 Static and Time-Series Approaches	17
2.3.2 Leveraging Alternative Data and Advanced Models	18
2.4 The Role of Agentic AI	19
2.4.1 Core Concepts and Architecture of Agentic AI	19
2.4.2 Agentic AI in Financial Decision-Making	20
2.5 Research Gaps and Justification	20
2.5.1 Limitations of Existing Frameworks	21
2.5.2 The Justification for an Agentic AI Solution	21
2.6 Analysis of Existing Algorithms	22
2.6.1 Traditional Predictive Models	22
2.6.2 Machine Learning Predictive Models	23
3 Methodology and Requirements Analysis	27
3.1 Methodology	27
3.1.1 System FlowChart	27
3.1.2 Data Science Cycle : Crisp-DM	30
3.1.3 Agentic Artificial Intelligence Cycle	33
3.1.4 Software Development Life Cycle	35
3.2 Requirement Gathering Techniques	38
3.2.1 Interview	38
3.2.2 Questionnaires	39
3.3 Functional and Non Functional Requirement	40

3.3.1 Functional Requirements	40
3.3.2 Non - Functional Requirements	42
3.4 Description of Dataset	43
3.4.1 Data Structure and Data Dictionary	43
3.5 Chapter Summary and Evaluation	44
4 System Design	47
4.1 System Architecture Design	47
4.1.1 User Interface - 1st Tier	47
4.1.2 Presentation Layer - 2nd tier	48
4.1.3 Application Layer - 2nd Tier	48
4.1.4 Domain Model Layer - 2nd Tier	48
4.1.5 Persistence Layer - 2nd Tier	49
4.1.6 Data Layer - 3rd Layer	49
4.1.7 Infrastructure	49
4.2 User Interface (UI) Design	49
4.3 Algorithm and Method Used	50
4.4 Chapter Summary and Evaluation	51
5 Implementation and Testing	53
5.1 Sub-section 1 Heading	53
5.1.1 Sub-subsection Heading	53
5.2 Sub-section 2 Heading	53
5.3 Sub-section 1 Heading	54
5.3.1 Sub-subsection Heading	54
5.4 Sub-section 1 Heading	54
5.4.1 Sub-subsection Heading	54
5.5 Chapter Summary and Evaluation	54
6 System Deployment	56
System Backup and Risk Management	56
On-site Setup	56
Training Procedure	56
Follow-up	56
Chapter Summary and Evaluation	56
7 Discussions and Conclusion	58
Summary	58
Achievements	58
Contributions	58
Limitations and Future Improvements	58
Issues and Solutions	58
References	60

Chapter 1

Introduction

1 Introduction

This chapter introduces the development of an Agentic AI-based Credit Modeling System designed to improve loan approval decisions in the banking sector. Focusing on salary prediction as the main feature, supported by loan interest and credit risk prediction, the system applies AI and machine learning to provide more accurate and adaptive assessments of repayment capacity. The chapter outlines the background, objectives, scope and contributions of the project, setting the foundation for its design and implementation.

1.1 Project Overview.

This Final Year Project focuses on the development of an Agentic AI-based credit modeling system for the banking and finance sector, aimed at enhancing decision-making in loan approval processes through predictive intelligence. Unlike conventional models that rely heavily on historical income records and rigid rule-based systems, the proposed system leverages Machine Learning (ML) and Agentic AI to make credit risk modeling more dynamic, adaptive and accurate in assessing repayment capacity.

The system is structured around four interconnected submodules: Salary Prediction, Loan Interest Prediction Based on Risk, Credit Risk Prediction and Retrieval-Augmented Generation (RAG) Modeling. Among these, the primary emphasis is placed on Salary Prediction, which estimates a customer's income trajectory using historical financial data and career-related indicators. This function not only supports more accurate loan approval decisions but also enables banks to evaluate long-term customer eligibility for credit limit increases, thereby improving trust and service quality. The second submodule, Loan Interest Prediction Based on Risk, dynamically adjusts loan interest rates according to a borrower's financial risk profile, helping institutions safeguard lending operations while maintaining profitability. The third submodule, Credit Risk Prediction, anticipates potential defaults, allowing banks to better manage non-performing loans and minimize financial losses. Finally, the RAG Modeling submodule enhances the system's adaptability by integrating external knowledge retrieval into the decision-making process, reducing errors from outdated data and ensuring more transparent, context-aware credit assessments.

By using synthetic financial data and deploying the prototype through an API, this project demonstrates how AI-driven tools can be practically integrated into credit evaluation workflows, ensuring loan approvals that are more accurate, fair and efficient, while laying the foundation for broader applications of Agentic AI in financial risk management.

1.2 Project Background.

The banking and finance industry is a cornerstone of economic growth, as it provides credit that supports business expansion, household investments and personal financial security. Extending credit carries inherent risks, requiring lenders to balance profitability against the likelihood of borrower default. Traditionally, credit evaluation has relied on manual reviews and rigid scoring systems built on fixed criteria such as income levels, repayment history and collateral. While effective in the past, these approaches often overlook the complexities of today's financial landscape, where shifting income patterns, diverse employment types and evolving consumer behaviors demand more adaptive assessment methods.

These challenges have become more pronounced with the digitalization of financial services, the expansion of gig economy employment and global disruptions like economic crises and pandemics, all of which create instability in income and repayment capacity. Additional obstacles such as data drift, fragmented records across institutions and inconsistent data quality further weaken traditional models. Meanwhile, regulators increasingly call for risk evaluation frameworks that are transparent, fair and adaptable to modern financial realities.

In response, credit risk modeling has begun leveraging artificial intelligence (AI) and machine learning (ML), which can process vast datasets, uncover subtle risk indicators and deliver more accurate predictions of borrower behavior. However, many conventional ML systems remain limited by their static nature, requiring ongoing retraining and close human supervision. Agentic AI offers a significant advancement by merging predictive analytics with autonomous reasoning and adaptive learning, allowing credit models to not only estimate default probabilities but also evolve in real time as new data becomes available.

Within this broader approach to credit risk modeling, salary prediction emerges as an essential subcomponent. Since repayment ability is closely linked to future income, forecasting salary progression enhances the accuracy of credit assessments. By applying Agentic AI to salary prediction, models can continuously refine income forecasts, offering financial institutions deeper insights into borrower stability. This integration strengthens loan decision-making, minimizes non-performing loans and fosters greater trust between banks and their clients.

1.2.1 Problem Statement

In the banking and financial sector, the loan approval process is one of the most critical services offered to customers because it determines whether an individual or a business can gain access to essential financial resources. Traditionally, banks have relied on manual assessments and rigid,

rule-based systems to evaluate loan applications. Loan officers would verify income documents, review repayment histories and use fixed scoring formulas to decide eligibility. While these methods have served their purpose for decades, they are far from perfect. Manual assessments are often time-consuming and labor-intensive and the decisions made may be influenced by human bias or simple human error. Rule-based systems, on the other hand, lack flexibility. They operate on predefined thresholds that rarely capture the complexities of modern financial behavior, leaving banks at risk of both rejecting reliable customers and approving risky borrowers (Hand & Henley, 1997).

As the demand for credit facilities continues to grow, the shortcomings of these traditional approaches have become even more apparent. Financial institutions today must process thousands of applications in a short period of time, yet the existing systems struggle to scale efficiently. More importantly, they lack the predictive power to assess future trends. Most loan evaluations still rely heavily on past records, without accounting for the possibility of income growth, changes in career trajectory or external economic factors (Abdou & Pointon, 2011). This means that some applicants who could repay their loans are unfairly rejected, while others who pose a high risk of default may still be approved. These inefficiencies not only hurt customer satisfaction but also expose banks to financial losses and reputational damage.

Besides that, data drift and data quality issues are two significant limitations of the existing system. Data drift happens when the data used by a model changes over time, but the system continues to operate on outdated patterns. In banking, this occurs because customer behavior, market conditions and the economy are always evolving, while many banks rely on legacy systems. For example, the rise of gig economy jobs, digital payments and sudden events such as pandemics or inflation can drastically change income stability. If models are not updated frequently, they may misclassify reliable borrowers as risky or approve customers who are likely to default. Such mistakes increase loan defaults, reduce customer trust and make banks less competitive compared to fintech companies that deploy more adaptive models.

Data quality issues are another persistent problem. Many banks still struggle with redundant, duplicate or inconsistent records due to weak data integration across branches and systems. For instance, a repayment record might be updated in one branch but not reflected in the central database, causing a borrower to be mislabeled as high-risk. Duplicate or incorrect entries may also inflate debt levels, leading to unfair rejections. Poor data quality not only harms customers but also creates operational inefficiencies and exposes banks to regulatory penalties.

Therefore, this is where Artificial Intelligence (AI) and Machine Learning (ML) are beginning to transform credit modeling. Unlike static rule-based systems, AI-driven models can analyze large volumes of financial data, detect hidden patterns and make predictions with higher accuracy. For example, machine learning algorithms can forecast a borrower's likelihood of income growth, detect repayment risks early and dynamically adjust to new data (Lessmann et al., 2015). The emergence of Agentic AI further enhances this capability by enabling systems to reason, learn and refine predictions autonomously, reducing reliance on manual interventions (Bharath Somu, 2024). In the context of credit evaluation, this enables banks to move away from rigid scoring systems toward adaptive, transparent and customer-centric decision-making.

This Final Year Project aims to address that gap by developing an Agentic AI-based credit modeling system with a particular focus on future salary prediction. By analyzing diverse customer attributes and financial indicators, the system seeks to forecast future salary and repayment capacity more accurately, thereby assisting banks in making fairer loan approval decisions. At the same time, it lays the groundwork for advanced modules such as loan interest prediction and good-payee versus bad-payee forecasting. In doing so, this project highlights how AI can deliver smarter, fairer and more efficient alternatives to traditional evaluation methods, enabling financial institutions to reduce risk while offering improved customer experiences.

1.3 Project Objective

- To develop an Agentic AI based credit modeling system that predicts a customer's future salary for assessing repayment capacity and supporting loan approval decisions.
- To integrate affordability insights from predicted salary into loan application processes, supporting more accurate credit limit adjustments and long-term customer relationship management.
- To implement the Agentic AI credit modeling system as an API, enabling financial institutions to seamlessly integrate and utilize the model's predictions within their existing loan approval platforms.

1.4 Scope of Project

The scope of this project is defined to ensure that the development of the Agentic AI-based credit modeling system remains focused, achievable and aligned with the research objectives.

The system will primarily emphasize future salary prediction as its core function. Since

repayment capacity is closely tied to income growth and stability, forecasting future salary provides a clear and measurable way to support loan approval decisions. By focusing on this specific objective, the project maintains clarity of direction while contributing practical value to the field of credit modeling.

1. Data Creation and Usage

The first challenge in developing financial AI systems is the limited availability of real banking datasets, as they are highly sensitive and protected by strict privacy and compliance regulations. To overcome this, the project adopts a hybrid data strategy by combining synthetic data generated with the Python Faker library and publicly available datasets from Kaggle. The Faker library is widely used to create realistic yet fictitious records, simulating key attributes such as demographic information, employment details, income history and repayment records. These synthetic features are complemented with Kaggle datasets, which provide structured financial and credit-related information that enhances the realism and variability of the training data. Python libraries such as pandas and NumPy are used to generate, organize and preprocess the data for modeling. Although synthetic and open-source datasets cannot capture the full complexity of proprietary banking records, this approach offers a safe, ethical and flexible environment for experimentation, while allowing new attributes to be added when needed.

2. Boundary of the Credit Model

The system is designed with clearly defined boundaries to keep the project practical and manageable. The credit modeling system will focus solely on predicting a customer's future salary as an indicator of repayment capacity. It will not attempt to replicate a complete credit scoring system, which in real banking environments often includes factors such as collateral, behavioral data, regulatory compliance and macroeconomic conditions. While these factors are important, they are outside the scope of this project due to time, resource and data constraints. Instead, the system aims to deliver reliable predictive insights on salary growth trends that can complement existing credit evaluation processes.

3. Artificial Intelligence and Agentic AI Implementation

The predictive model will be developed using Python-based machine learning techniques, with a particular focus on supervised learning algorithms suited for regression tasks such as salary forecasting. Libraries such as scikit-learn, TensorFlow and PyTorch will be used to build, train and evaluate the models on the synthetic dataset. Importantly, Agentic AI principles will be incorporated into the development of the system. These features will enable the model to adapt to new data, refine predictions over time and incorporate elements of autonomous

decision-making. For example, the system will include mechanisms such as automated retraining triggers and incremental updates. These adaptive and reasoning-driven features are a central innovation of the project and are intended to showcase how Agentic AI can enhance predictive accuracy in credit modeling.

4. API Implementation

To demonstrate practical usability, the credit modeling system will be implemented as a Python-based Application Programming Interface (API). Using frameworks such as Flask or FastAPI, the model's predictions will be exposed through endpoints that external systems can call. This approach makes it possible, in principle, for financial institutions to integrate the model into their loan approval workflows without overhauling their existing infrastructure. The API will be presented as a functional prototype to showcase how the system's predictions can be accessed and utilized programmatically. However, full-scale integration with live financial institutions, compliance with banking regulations and deployment in production environments will remain outside the scope of this project.

In summary, this project will focus on the design and development of an Agentic AI-based credit modeling prototype, with future salary prediction as its sole core feature. The system will make use of synthetic data generated in Python, processed and modeled using Machine Learning and Agentic AI techniques and made accessible through a Python-based API. The project is limited to building and testing a prototype that supports loan approval decisions by providing insights into repayment capacity. Full-scale production deployment, real-time banking integration and compliance with financial regulations are beyond the scope of this research but represent directions for future development.

1.5 Advantages and Contribution

The development of an Agentic AI based salary prediction module for credit modeling addresses both commercial and social needs while aligning with national and global development goals. By forecasting a customer's future income, the module helps banks and financial institutions make better-informed loan decisions, ensuring that approved loans are realistic and sustainable. This improves repayment capacity assessment, reduces the likelihood of defaults and strengthens trust between customers and financial institutions.

Commercialisation Value

The proposed salary prediction module carries significant commercialisation value, particularly within the banking, credit and fintech industries. It can be adopted by financial institutions as an integral part of loan approval and credit limit evaluation processes. By helping banks identify long-term customers who are eligible for credit limit increases, the system not only strengthens risk management but also enhances customer loyalty and retention. At the same time, it reduces credit risk through more accurate assessments of repayment capacity, ensuring that loans are extended responsibly and sustainably. Furthermore, by automating parts of the loan evaluation process, the module minimizes reliance on manual reviews, reduces operational costs and improves overall efficiency, making it a practical and cost-effective solution for modern financial institutions.

Social Impact

The salary prediction module contributes to society by promoting financial inclusion and responsible lending practices. By forecasting future income growth, the system enables banks to offer fairer access to loans for individuals who may have stable earning potential but limited credit history. This reduces the financial barriers faced by young professionals, entrepreneurs and underserved groups, giving them opportunities to access credit and improve their livelihoods. At the same time, by preventing irresponsible lending and potential over-borrowing, the system safeguards customers from financial distress and fosters greater trust in financial institutions. Beyond individual benefits, the project contributes to a healthier credit ecosystem where both borrowers and lenders can make informed, data-driven decisions.

Technological and Institutional Value

From a technological perspective, the salary prediction module demonstrates how advanced AI and machine learning techniques can be effectively applied to solve real-world financial challenges. By integrating predictive analytics into existing loan evaluation systems, the module strengthens the technological capabilities of banks, credit agencies and fintech platforms. On an institutional level, it provides a strong foundation for further innovation, such as integrating the module with broader credit scoring systems, personalized financial advisory services or even national credit bureaus. This not only enhances the digital transformation of financial institutions but also exposes industry practitioners and future professionals to cutting-edge technologies, supporting Malaysia's push toward a digitally skilled workforce and resilient financial infrastructure.

1.5.1 Alignment with Sustainable Development Goals (SDGs) and Malaysia Madani

SDG 8 – Decent Work and Economic Growth

This project supports SDG 8 by promoting responsible lending practices that reflect a borrower's real earning capacity. By predicting future salaries, banks can make decisions that reduce the risk of over-indebtedness while still supporting individuals with the financial resources they need to grow. This creates a healthier balance between protecting borrowers and ensuring banks can support entrepreneurship, education and career development. In the bigger picture, such practices help stabilize the credit market, reduce loan defaults and drive long-term economic growth.

SDG 9 – Industry, Innovation and Infrastructure

The salary prediction module directly contributes to SDG 9 by modernizing financial infrastructure through the integration of AI and data-driven intelligence. Traditional methods often rely on static rules and manual evaluations, but this project introduces adaptive, predictive analytics that improve efficiency and transparency in credit decisions. Beyond improving current processes, it also opens pathways for innovation in financial services, such as personalized credit packages or predictive financial planning tools. This helps strengthen the competitiveness and resilience of Malaysia's financial industry in line with global innovation trends.

SDG 10 – Reduced Inequalities

Financial exclusion is a pressing challenge for individuals without strong credit histories, such as young graduates, freelancers and gig economy workers. By forecasting income growth potential, this project helps these groups access fairer loan opportunities, rather than being unfairly excluded by rigid eligibility criteria. This aligns with SDG 10 by reducing inequalities in financial access, fostering inclusivity and ensuring that opportunities are available to all who demonstrate repayment potential, not just those with established credit records.

Alignment with Malaysia MADANI (Digital MADANI – Teknologi Digital dan Inovasi)

This project also reflects the aspirations of Malaysia MADANI, particularly under the theme of Digital MADANI (Teknologi Digital dan Inovasi). By embedding AI-driven intelligence into the banking sector, it promotes innovation, transparency and inclusivity in financial services. The salary prediction module empowers institutions to adopt smarter, data-driven approaches, reducing human bias and ensuring fairer loan approvals. At the same time, it prepares Malaysia's workforce and financial institutions for the digital economy by fostering digital literacy and embracing technological transformation. In line with the MADANI vision, this project not only strengthens the financial sector but also contributes to a more progressive, equitable and sustainable society.

1.6 Project Plan & Gantt Chart

1.6.1 Project Plan

Table 1.1 Project Plan

Start Date	Project Phases	Activities	Duration
26/09/2025	Proposal Phase	Submit finalized project proposal for approval	1 day
26/09/2025	Project I Submission	Successfully submit Project I document	1 day
27/09/2025	System Analysis, Planning & Data Collection	Define system scope, outline features and collect relevant datasets	7 days
04/10/2025	Design & Development	Complete system architecture design, database design and develop main modules	27 days
31/10/2025	Testing Phase	Conduct unit testing, integration testing and refine modules	7 days
07/11/2025	Implementation & Deployment	Deploy and integrate the developed system with the main environment	3 days
10/11/2025	Test Plan Preparation	Submit a comprehensive test plan and test cases for evaluation	4 days
14/11/2025	System Preview	Demonstrate system functionality and progress to supervisor for feedback	7 days
21/11/2025	Final System Testing	Conduct joint testing with supervisor and moderator	7 days
28/11/2025	Draft Report Writing	Prepare and write draft FYP report	14 days
13/12/2025	Final Report Writing	Complete and refine final report (Chapters 1–6)	7 days

19/12/2025	Final Report Submission	Submit final FYP report	1 days
------------	-------------------------	-------------------------	--------

1.6.2 Gantt Chart

1.2 Table of Gantt Chart

Activity	September 2025		October 2025					November 2025				December 2025		
	W4	W5	W1	W2	W3	W4	W5	W1	W2	W3	W4	W1	W2	W3
Proposal Submission														
Project I Submission														
System Analysis, Planning and Data Collection														
Design and Development														
Testing Phase														
Implementation and Deployment														
Test Plan Preparation														
System Preview														
Final System Testing														
Draft Report Writing														
Final Report Writing														
Final Report Submission														

1.7 Project Team and Organisation

1.7.1 Organisation Chart



Figure 1.6 : Figure of Organisation Chart

The organization chart outlines the structure of our project team and the division of responsibilities under the guidance of Assoc. Prof. Ts. Dr. Yu Yong Poh, our project supervisor. He provides continuous academic guidance, monitors progress and ensures that our objectives and methodologies align with best practices in financial technology research.

Our team is divided into four submodules, each focusing on a different aspect of the credit modeling system. Chan Jin Wei leads the Credit Module: Risk Modeling, responsible for identifying, analyzing and quantifying potential risks associated with borrowers through predictive models that assess default probability and creditworthiness. Lai Yoke Yau manages the Credit Module: Adaptive Interest Rate Modeling, which dynamically predicts and adjusts interest rates based on identified risk factors to balance profitability with fairness. I, Ethel Ng Yi Yan, am responsible for the Credit Module: Salary Prediction Modeling, where I develop models that forecast future income levels using machine learning techniques. This strengthens the system's ability to evaluate repayment capacity by providing more reliable salary projections. Gan She Ming leads the RAG Module: Retrieval-Augmented Generation Modeling, which integrates knowledge retrieval into the system, making credit assessments more adaptive, transparent and context-aware.

Together, we work collaboratively to ensure that each module integrates seamlessly into a cohesive system. This division of labor not only improves efficiency but also allows each of us to specialize in a critical area, resulting in a more robust and innovative Agentic AI-based credit modeling framework.

1.8 Chapter Summary and Evaluation

This chapter introduced the motivation and foundation for developing an Agentic AI-based credit modeling system to address the shortcomings of traditional loan approval methods, which are manual, rigid and prone to bias, data drift and inefficiencies. The project centers on a salary prediction model that forecasts future income to assess repayment capacity, supported by secondary objectives such as integrating affordability insights, deploying a prototype API while enabling adaptive retraining through Agentic AI.

The scope focuses on salary prediction using synthetic datasets generated with Python's Faker library, applying regression-based ML techniques enhanced with Agentic AI for reasoning and adaptation. By demonstrating practical integration into banking workflows, the system offers academic, technological, commercial and social value, while reducing financial risk, promoting fairer access to credit and supporting Malaysia's digital transformation while laying the groundwork for future applications in financial risk management.

Chapter 2

Literature Review

2 Literature Review

2.1 Introduction

The integration of artificial intelligence (AI) into credit assessment processes has transformed how financial institutions assess borrower reliability and repayment capacity. Traditional credit scoring frameworks, such as logistic regression or rule-based risk assessment have been widely used for decades. However, these models often fail to incorporate dynamic labor market conditions, evolving income patterns and unstructured data that can significantly affect an individual's earning potential (Khandani et al., 2010). Advances in machine learning have further improved credit scoring, enabling the use of classification algorithms that outperform many traditional approaches in predictive accuracy (Lessmann et al., 2015).

A new frontier in this evolution is the emergence of Agentic AI, which goes beyond static prediction to enable autonomous reasoning, planning and adaptation. Unlike conventional AI models that provide a single output, agentic systems can continuously learn from new data, use external tools (like APIs and databases) and refine their strategies to achieve a specific goal (Park et al., 2023). This capability is particularly transformative for loan evaluation, as it allows for a more dynamic assessment of a borrower's financial health, rather than relying on a fixed snapshot of their past.

This project introduces a novel approach by developing an Agentic AI-based system for future salary prediction, a critical component of assessing repayment capacity. By forecasting a customer's potential income growth, the system provides a forward-looking metric that complements traditional credit history. This approach not only enhances the accuracy and adaptability of loan approval decisions but also addresses the limitations of legacy systems, such as their susceptibility to data drift and an inability to account for evolving financial behavior.

This chapter reviews four interlinked areas central to the proposed work, which are the landscape of credit assessment, the salary prediction or income forecasting, the role of agentic AI as well as the research gaps and justifications.

2.2 The landscape of Credit Assessment

2.2.1 Traditional Credit Scoring and Its Limitations

Traditional credit assessment models are inherently backward-looking, relying on historical financial data such as repayment records, outstanding debt and the duration of credit history to determine a borrower's creditworthiness. These models, exemplified by the widely-used FICO score, were designed for a more stable economic environment, operating on the linear assumption that a consistent increase or decrease in a single variable directly impacts the final score (Thomas, 2009). This approach, while providing a clear and auditable method for risk assessment, oversimplifies the complex, non-linear relationships that govern modern borrower behavior. A significant limitation is the inherent rigidity of these systems; they cannot adapt to sudden macroeconomic shifts, as was evident during the 2008 financial crisis when models built on pre-crisis data proved insufficient for evaluating new credit risks (Khandani et al., 2010). Furthermore, traditional models suffer from a fundamental problem of financial exclusion, as they are unable to fairly evaluate individuals with a limited or "thin" credit file, such as young professionals, immigrants or those in the gig economy (Brevoort et al., 2016). These populations are often deemed high-risk or unscorable, denying them access to credit despite having a stable income and strong financial profile from non-traditional sources. This exclusionary nature not only restricts access to credit but also perpetuates a cycle of financial marginalization, a well-documented issue in economic literature (Fairness and Machine Learning, 2023).

The core of traditional credit risk modeling is its reliance on historical, structured data, which creates a significant vulnerability known as concept drift. This occurs when the statistical properties of the target variable like the loan default may change over time, rendering a static model's predictions less accurate (Goodell, 2020). For instance, an economic recession, a pandemic or a new government policy can fundamentally alter the relationship between a borrower's income and their ability to repay a loan, causing the model's performance to degrade without warning (Aldasoro et al., 2020). Beyond this, the labor-intensive process of manually updating and re-validating these models makes them ill-equipped to handle the speed and volume of modern financial transactions. Even with robust governance frameworks, the time-lag between a significant market event and the re-calibration of a static credit model can expose a financial institution to considerable risk. Therefore, a reliance on these legacy systems, despite their established interpretability, leaves a critical gap in the financial sector's ability to conduct agile and inclusive risk management.

2.2.2 Transition to Data-Driven and AI-Based Models

In response to the limitations of traditional methods, financial institutions have rapidly adopted machine learning (ML) models. The advent of big data and powerful computing resources has enabled the use of algorithms like Random Forests, Gradient Boosting Machines and Support Vector Machines (SVMs), which are capable of identifying intricate patterns and non-linear correlations in vast, high-dimensional datasets (Lessmann et al., 2015). These models have been proven to significantly outperform their traditional counterparts in predictive accuracy, particularly for fraud detection and algorithmic trading, where speed and precision are paramount (Casolaro et al., 2023). By moving beyond simple linear relationships, ML-based credit scoring has provided banks with a more comprehensive view of risk, allowing them to segment customers more effectively and tailor credit products to specific risk profiles. The success of these models has been foundational to the rise of fintech and the introduction of new financial products that rely on sophisticated data analytics (Smythos, 2025).

However, the widespread adoption of these advanced models has introduced a new set of challenges, most notably the "black box" problem. The complexity of models like deep learning networks makes it difficult, if not impossible, for a human to understand the precise logic behind a decision. This lack of explainability is a major barrier to regulatory compliance, as oversight bodies require financial institutions to provide clear, auditable justifications for credit decisions to prevent discrimination and ensure fairness (Fairness and Machine Learning, 2023; Wachter et al., 2017). While Explainable AI (XAI) techniques like SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations) have emerged to provide some insight into model behavior, they do not fully solve the issue of accountability. Furthermore, these models, like their traditional predecessors, still require a "human-in-the-loop" for continuous monitoring, retraining and governance to mitigate issues like model drift. This dependency on manual oversight limits the scalability and real-time adaptability of even the most sophisticated ML frameworks, underscoring the need for a more autonomous and resilient solution.

2.3 Salary Prediction / Income Forecasting

2.3.1 Static and Time-Series Approaches

Early models for income prediction were deeply rooted in econometrics, with the Mincerian earnings function serving as a foundational benchmark. These static models estimated income as a function of an individual's years of schooling and work experience, providing a simple, interpretable framework

that was widely used in labor economics (Mincer, 1974). However, this approach is inherently limited to a narrow set of variables and cannot account for the vast complexity of an individual's career path, which is influenced by job market dynamics, sectoral shifts and personal skill development. For example, a Mincerian model would fail to distinguish between a software developer's salary in a high-growth tech hub versus a legacy industry, as it does not incorporate real-time market data or unstructured information about specific skills (Samuel Iorhemen Ayua et al., 2023). More recently, time-series analysis has been employed to model how income evolves over a career, using historical data to capture trends and seasonality. While an improvement, these models are often unable to predict the impact of unforeseen events like a pandemic or a sudden industry downturn, which can dramatically alter an individual's earning trajectory, a fact highlighted by recent economic studies (Goodell, 2020).

The reliance on static and historical data in these models makes them susceptible to forecasting errors in a volatile labor market. For instance, a model trained on pre-2020 data would be ill-equipped to accurately predict the income of a freelance writer or a rideshare driver, as these new work models lack consistent historical data and are subject to market fluctuations that defy traditional linear forecasting (Jagtiani & Lemieux, 2019). Furthermore, these models often rely on self-reported data, which can be inaccurate or incomplete. The lack of a mechanism to verify and update income information in real-time limits their utility for high-stakes financial applications like credit underwriting. The predictive power of these models, therefore, hinges on a set of assumptions that often do not hold true in modern, dynamic employment environments. This justifies the need for an approach that can not only ingest heterogeneous data but also adapt to a constantly changing world of work.

2.3.2 Leveraging Alternative Data and Advanced Models

To overcome the limitations of traditional models, researchers have turned to alternative data sources and advanced machine learning techniques. This approach involves leveraging unstructured data from a variety of sources, including professional resumes, job market listings and even social media profiles, to build a more holistic picture of an individual's earning potential (Gomez-Cravioto et al., 2022). For instance, an analysis of keywords on a LinkedIn profile or GitHub repository can provide granular insights into a person's specialized skills, which are often a better predictor of salary than a generic job title. Advanced models, particularly deep learning networks, are uniquely suited to process this heterogeneous data, as they can learn complex representations and patterns that would be missed by simpler statistical models (Joshi et al., 2022). This ability to synthesize diverse data points allows

for a more personalized and accurate income forecast, especially for individuals with non-linear career paths or those in emerging fields where historical data is scarce.

This transition to alternative data also serves a critical social purpose: promoting financial inclusion. By incorporating information such as utility payments, rental history and digital transaction footprints, these models can effectively assess the creditworthiness of individuals who have been traditionally excluded from the credit system due to a lack of formal credit history (Jagtiani & Lemieux, 2019; MDPI, 2022). This not only expands the pool of eligible borrowers but also provides a fairer, more comprehensive evaluation of their financial responsibility. However, the use of this data is not without its risks. The models' effectiveness is highly dependent on the quality and representativeness of the data and there is a significant risk of inadvertently amplifying pre-existing biases related to socioeconomic status, geography or even race (Carter, n.d.). Therefore, while these advanced approaches offer immense promise, they must be developed with a strong commitment to ethical guidelines and rigorous testing to ensure fairness and prevent unintended harm.

2.4 The Role of Agentic AI

2.4.1 Core Concepts and Architecture of Agentic AI

Agentic AI distinguishes itself from traditional AI by its ability to engage in autonomous reasoning, planning and self-correction to achieve a defined goal (Park et al., 2023). Unlike a conventional machine learning model that acts as a passive function, an agentic system is an active entity that perceives its environment, reasons about its next steps and performs actions without human intervention. The core architectural components of an agentic system include a planner to break down complex tasks into manageable sub-tasks, a memory module (both short-term and long-term) to retain and retrieve context and a tool-use capability that allows it to interact with external systems like databases, APIs and other software applications. This architecture enables a continuous loop of perception, thought and action, which is a significant departure from the static input-output mapping of a typical predictive model. For a credit assessment system, this means the agent can not only predict a risk score but can also plan a strategy to retrieve additional data, interact with a user or initiate a document-processing task to fulfill its goal of a comprehensive credit evaluation.

This autonomous and adaptive nature is fundamentally what makes Agentic AI a revolutionary technology. It allows a system to operate in a dynamic, real-world environment where conditions are constantly changing. While a static ML model would require a human to manually trigger a retraining process to address concept drift, an agentic system can be designed to detect and respond to these changes on its own, ensuring its predictions remain relevant and accurate over time (Scheffler, 2025;

Goodell, 2020). The ability to use external tools is equally transformative, as it allows the agent to pull in real-time market data, regulatory updates or alternative credit indicators from disparate sources and synthesize them into its decision-making process. This dynamic, proactive capability makes agentic systems particularly well-suited for high-stakes financial applications where decisions must be both precise and agile.

2.4.2 Agentic AI in Financial Decision-Making

The application of Agentic AI in financial services extends far beyond simple prediction. These intelligent agents can be deployed across a range of high-stakes tasks, from algorithmic trading and fraud detection to personalized financial advice (Smythos, 2025). For a credit assessment system, an agentic framework can continuously monitor a borrower's financial activities, adjusting their risk profile in real-time based on new data. For example, if a borrower experiences an unexpected job change, the agent can autonomously retrieve new employment and income data and instantly adjust the credit score, without a manual review process. This dynamic monitoring capability reduces reliance on outdated, snapshot-based assessments and helps banks respond to economic shocks or unexpected borrower behavior with agility. The Agentic AI system can also engage in complex, multi-step tasks, such as cross-referencing a loan application against multiple regulatory databases and generating an auditable report for a compliance officer, a process that would otherwise take days of manual labor.

Furthermore, Agentic AI enhances the governance of financial models through its inherent ability to generate traceable, evidence-based justifications. By leveraging tools like Retrieval-Augmented Generation (RAG), an agent can not only make a decision but also provide a transparent, evidence-based explanation by referencing the specific data points, documents or regulatory guidelines that influenced its choice (Lewis et al., 2020). This provides an unprecedented level of transparency and auditability, bridging the gap between machine intelligence and regulatory compliance. The integration of structured (transaction records) and unstructured (KYC documents, legal guidelines) data gives it a strategic advantage in regulated financial domains. By combining prediction with reasoning and action, Agentic AI moves financial services from being reactive to proactive, ensuring that decisions are not only accurate but also robust, compliant and adaptable to a constantly evolving landscape.

2.5 Research Gaps and Justification

While significant progress has been made in applying AI to credit risk and salary prediction, a critical research gap persists. The majority of existing models are either static or rely on a "human-in-the-loop" to manage their performance, which limits their efficiency and long-term adaptability. The financial industry's need for real-time, explainable and scalable systems remains largely unaddressed by current technology.

2.5.1 Limitations of Existing Frameworks

The core limitation of most existing AI credit assessment systems is their lack of autonomy and adaptability. They cannot proactively respond to changes in macroeconomic conditions, a borrower's personal circumstances or the regulatory environment without manual retraining. This static nature makes them highly vulnerable to concept drift, a phenomenon where the relationship between input features and the target variable changes over time, causing a model's performance to degrade (Goodell, 2020). While techniques for monitoring and mitigating this drift exist, they are often reactive and require a continuous, labor-intensive cycle of human intervention, model retraining and re-deployment. Furthermore, while the accuracy of complex models has improved, their opaque nature remains a significant regulatory and ethical hurdle. The "black box" problem prevents financial institutions from providing clear, understandable justifications for why a loan was approved or denied, which is a fundamental requirement for fair lending practices (Fairness and Machine Learning, 2023; Rudin, 2019). The current state of AI in credit assessment, therefore, presents a trade-off between predictive accuracy and interpretability.

Moreover, the existing literature on salary and income prediction has largely focused on forecasting at a single point in time or on a static career path, often failing to account for the dynamic, non-linear nature of modern employment. A significant portion of the workforce now participates in the gig economy, where income streams are inconsistent and defy traditional forecasting models (Jagtiani & Lemieux, 2019). A lack of a unified, adaptable framework that can integrate real-time income data with forward-looking projections is a major gap. Existing models are siloed, meaning that a credit risk model and a salary prediction model are developed and managed separately, limiting their synergistic potential. The challenge is not just to build a more accurate predictive model, but to create a holistic system that can continuously learn, adapt and reason across multiple data sources to provide a truly comprehensive and resilient assessment of a borrower's financial health.

2.5.2 The Justification for an Agentic AI Solution

This project directly addresses these gaps by proposing an Agentic AI-based system. An agentic framework is uniquely positioned to handle the challenges of dynamism, data drift and explainability that plague existing models. By integrating its predictive engine with reasoning and tool-use capabilities, the system can autonomously adapt to new data, self-correct its predictions and generate evidence-based justifications for its decisions (Lewis et al., 2020). Unlike traditional models that require a human to intervene when an anomaly is detected, an agentic system can be programmed to proactively respond, such as by fetching a user's updated employment data or flagging a suspicious transaction for further review, all without manual intervention. This moves the system from a reactive to a proactive state, significantly enhancing its efficiency and accuracy in a fast-paced financial environment.

The proposed system also provides a robust solution to the "black box" problem. By leveraging the agent's memory and reasoning capabilities, a clear audit trail can be created for every decision. The agent can provide a comprehensive report that details the specific data points it considered, the external sources it accessed and the logical steps it took to arrive at a conclusion. This level of transparency not only meets regulatory requirements but also builds trust with consumers, who are increasingly wary of automated financial decisions. The goal is to create a lending tool that is not only more accurate and efficient but also more responsible and equitable, thereby promoting financial inclusion and trust in automated financial systems. The development of such a system represents a critical step toward the future of credit assessment, where intelligence, adaptability and accountability are seamlessly integrated.

2.6 Analysis of Existing Algorithms

2.6.1 Traditional Predictive Models

Table 2.1 Traditional Predictive Models

Models	Linear Regression	Logistic Regression	Time Series Model
Supporting Features	Assumes a linear relationship between input and output variables. Data points include continuous values like income, outstanding debt and years of credit history.	Uses both continuous (e.g., income, age) and categorical data (e.g., home ownership, marital status) to predict a binary outcome.	Relies on historical, sequential data (e.g., monthly income, quarterly revenue) to identify trends, seasonality and cycles.
Advantages	The coefficients are easy to understand and	Simplicity and Interpretability:	Pattern Recognition: Excellent at

	can be used to explain the influence of each variable on the prediction	Provides a clear probability score (e.g., 0.8 probability of default) that is easily understood by human analysts and regulators	forecasting a variable's future value based on its own past behavior. Useful for forecasting recurring income streams
Limitations	It assumes a linear relationship, which often doesn't hold true in complex financial data. Limited Scope: Can't handle categorical data well without complex transformations.	Linearity: Assumes a linear relationship between input features and the log-odds of the outcome, which may not capture complex financial behavior	External Factors: Struggles to incorporate external, unpredictable events like a sudden job loss or economic recession. Cannot easily integrate non-time-series data
Remarks	A foundational model, best suited for simple financial models where transparency is a primary concern.	A standard for credit risk assessment due to its interpretability and compliance with regulatory requirements.	Ideal for forecasting a single, consistent financial metric over time, but limited in its ability to handle external variables.

2.6.2 Machine Learning Predictive Models

Table 2.2 Machine Learning Predictive Models

Models	Random Forest	Gradient Boosting Machine	Neural Network	Support Vector Machine
Supporting Features	An ensemble model that combines the predictions of multiple decision trees. Features include a mix of structured data, such as income, age and credit score.	An ensemble of decision trees built sequentially, with each new tree correcting the errors of the previous one.	Uses a network of interconnected nodes (neurons) to learn complex patterns from large, heterogeneous datasets, including unstructured text from resumes and social media	Works by finding the optimal hyperplane that separates data points into different classes.
Advantages	High Accuracy: Reduces overfitting and improves predictive power by aggregating multiple models.	State-of-the-Art Performance: Often wins data science competitions due to its high accuracy and	Superior Non-linearity: Excellent at modeling complex, non-linear relationships that	Effective in High Dimensions: Works well with datasets that have many features, even with a relatively small

	Feature Importance: Can rank features by their importance in the final prediction	efficiency. Can handle missing data and complex feature interactions automatically	traditional models can't. Adaptability: Can be trained to handle new and unstructured data sources, such as textual data or images.	number of data points. Overfitting Control: The margin maximization principle helps prevent overfitting
Limitations	Lower Interpretability: The final "black box" nature of the model makes it harder to explain a specific decision, which is a major regulatory concern	"Black Box" Nature: Like Random Forest, its high complexity makes it difficult to interpret individual predictions. Highly sensitive to noisy data, which can lead to overfitting if not properly tuned.	Extreme "Black Box" Problem: The multi-layered architecture makes it very difficult to interpret how a decision was reached, posing a significant regulatory hurdle . High Computational Cost: Requires significant computing power and large datasets for training.	Computational Complexity: Can be slow to train on large datasets. Kernel Selection: The choice of kernel function is crucial and can significantly impact performance.
Remarks	A powerful and widely-used model for credit scoring due to its high accuracy and robustness.	A top-tier choice for high-stakes predictive tasks, but requires careful tuning and validation.	Best suited for modeling unstructured, high-dimensional data, where predictive accuracy is prioritized over immediate interpretability.	A robust model for credit scoring, offering a good balance between performance and theoretical rigor.

Chapter 3

Methodology and Requirements Analysis

3 Methodology and Requirements Analysis

This chapter introduces the methodology and requirements analysis for the salary prediction module within a credit risk modeling system. It outlines the complete process from data collection and preprocessing to model training, evaluation and deployment, guided by frameworks such as CRISP-DM, the Agentic AI Cycle and the Software Development Life Cycle (SDLC). The chapter also highlights both functional and non-functional requirements, alongside requirement gathering techniques like interviews and questionnaires, to ensure the system is accurate, secure and user-friendly. By integrating structured methodologies with adaptive AI capabilities, the chapter emphasizes the importance of technical rigor and practical relevance in supporting fair, data-driven loan decisions.

3.1 Methodology

Methodology is the structured approach or framework used to conduct research or develop a system. It defines the steps, techniques and processes to collect, analyze and interpret data or to design, build and evaluate a solution. In essence, it provides a clear roadmap to ensure the work is systematic, reliable and aligned with the project's objectives.

3.1.1 System FlowChart

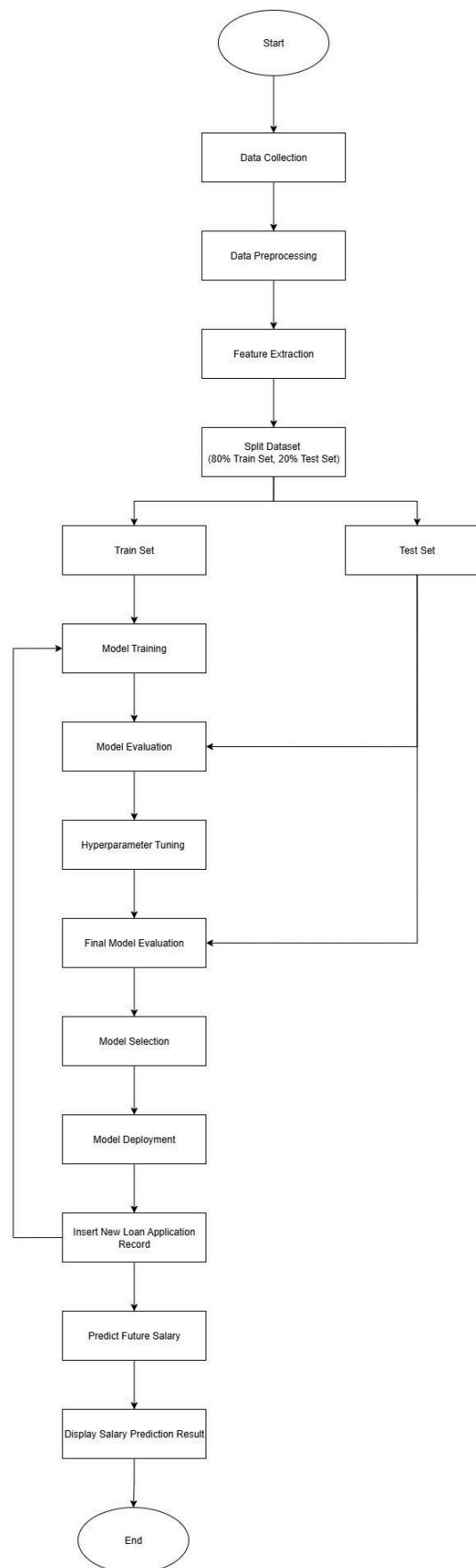


Figure 3.1 System Flowchart of Salary Prediction Module in Credit Risk Modeling System

The figure 3.1 above is about a flowchart which illustrates the complete workflow of the salary prediction module in the context of loan applications. Firstly, the process starts with data collection, where relevant financial and employment data is gathered from reliable sources such as bank records, salary slips or verified datasets. Once collected, the data will undergo preprocessing, which is a crucial step to ensure quality and consistency. During this stage, data cleaning is carried out to handle missing values, remove duplicates and errors corrections in the dataset so that the model does not learn from faulty information. Next, data normalization is applied to scale numerical features into a standard range, usually between 0 and 1, to avoid bias towards larger-valued features. Finally, data transformation converts raw data into a more suitable format, such as encoding categorical variables into numerical values or applying logarithmic transformations to reduce skewness, which improves model accuracy.

After data preprocessing, feature extraction is performed, where only the most relevant features that strongly affect salary prediction are selected. This step reduces unnecessary noise in the dataset and enhances computational efficiency by focusing on meaningful variables like years of experience, education level, job sector and employment history. Once the key features are identified, the dataset is divided into two parts: an 80% training set and a 20% test set. The training set is used for building and optimizing the prediction model, while the test set is reserved to measure how well the model generalizes to new, unseen data.

It continues with model training, evaluation and hyperparameter tuning, ensuring the system produces accurate predictions with the use of the train set. Model training, evaluation and hyperparameter tuning are key steps in developing a reliable salary prediction system. Model training is the process where the machine learning algorithm learns patterns and relationships from the training set (80% of the dataset). During this step, the model is fed with input features such as education, experience and job sector, along with the corresponding salary labels. The algorithm adjusts its internal parameters, such as the weights and biases to minimize the error between the predicted and actual salaries. Training continues until the model reaches an acceptable level of performance or until a stopping condition, which may be a set number of iterations or convergence is met.

After training, the model undergoes evaluation to check its effectiveness. This is done using the reserved test set (20% of the dataset), which the model has never trained before. Common metrics such as Mean Squared Error (MSE), Root Mean Squared Error (RMSE) or R^2 score (coefficient of determination) are used to measure how accurately the model predicts unseen data. If the model performs well on the training set but poorly on the test set, it indicates overfitting, meaning the model

has memorized patterns instead of learning general rules. On the other hand, if the performance is weak on both sets, the model may be underfitting, meaning it has not captured enough complexity of the data. Evaluation ensures that the model generalizes well to real-world applications. To further improve performance, hyperparameter tuning is performed. Hyperparameters are external settings that control the model's learning process, such as the learning rate in neural networks, the number of decision trees in Random Forest or the kernel type in Support Vector Machines. Unlike internal parameters, hyperparameters are not learned automatically and must be optimized manually or with systematic methods. Techniques such as Grid Search, Random Search or more advanced methods like Bayesian Optimization are commonly used to find the best hyperparameter values. By tuning these hyperparameters, the system improves prediction accuracy and ensures stability, resulting in a final model that can be confidently deployed for salary prediction in loan applications.

The best-performing model is then deployed into a real-world environment where new loan applications can be processed. At this stage, the applicant's data is inserted as a new loan application record and the system uses the trained model to predict the applicant's future salary. The prediction is then presented as the salary prediction result, which supports loan officers in making fair and data-driven decisions regarding loan approval. The record will then be inserted into the dataset and the model will be retrained to maintain efficiency and accuracy. This ensures transparency, reduces risks for the bank and improves accessibility for applicants who might otherwise face bias in traditional loan assessment methods.

3.1.2 Data Science Cycle : Crisp-DM

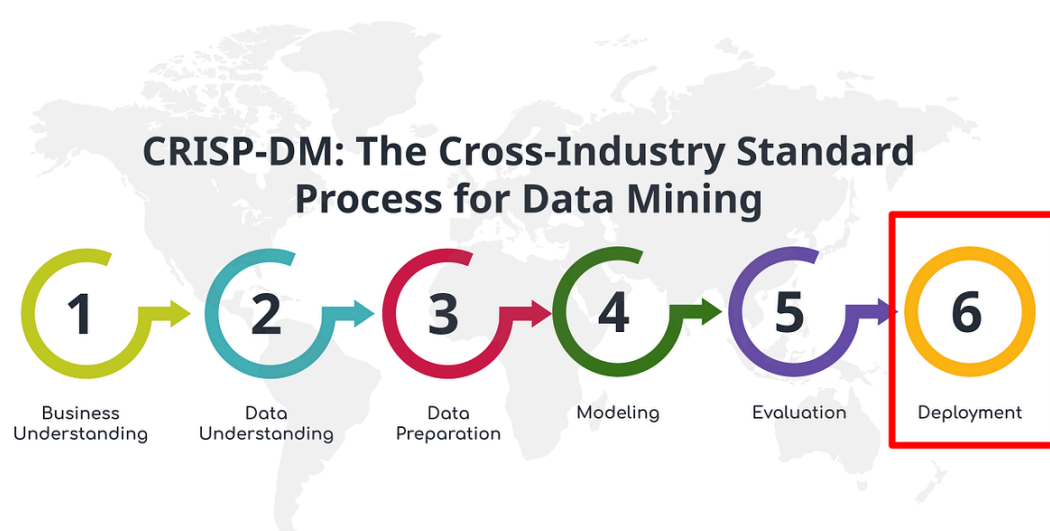


Figure 3.2 Process of Crisp-DM

CRISP-DM (Cross-Industry Standard Process for Data Mining) is a widely adopted methodology that provides a structured approach to data mining and machine learning projects. It is industry-neutral, meaning it can be applied across fields. This framework is flexible and iterative, allowing teams to revisit earlier steps as new insights emerge, rather than being confined to a linear process. One of CRISP-DM's key strengths is that it bridges the gap between technical processes and business objectives, ensuring results are both accurate and practically useful. By standardizing the workflow, it improves collaboration between data scientists and business experts, reducing miscommunication and aligning outcomes with organizational goals.

The CRISP-DM framework is divided into six phases, which are:

1. Business Understanding

The first phase of CRISP-DM emphasizes establishing a deep understanding of the business problem before any technical work begins. This involves identifying the goals, constraints and success criteria of the project from a business perspective. In the context of a salary prediction module used in the banking industry, the primary goal is to assist financial institutions in assessing a loan applicant's repayment ability by predicting their expected income based on available data. This helps loan officers make better-informed lending decisions, especially when applicants lack formal income documentation. Business understanding also involves analyzing the operational environment in which the solution will be deployed, identifying stakeholders' needs and clarifying how the model's results will influence decisions. Risks and limitations, such as data privacy regulations, bias in prediction, regulatory compliance and financial risk, are also considered at this stage. Ultimately, this phase ensures that the project's direction is grounded in delivering real value to the business rather than focusing solely on technical outcomes.

2. Data Understanding

After clarifying the business objectives, the next step is to gain familiarity with the data available. This phase begins with data collection, where datasets are gathered from multiple sources, such as customer application forms, credit reports, employment records and external income verification APIs. The goal is not only to gather data but to assess its quality, consistency and completeness. Exploratory Data Analysis (EDA) techniques, such as generating summary statistics, creating data visualizations or examining correlations are used to detect missing values, anomalies or unusual distributions. For example, when predicting applicant salaries, analysts would check whether critical features such as job title, industry,

education level, years of experience and location are present and reliable. By the end of this phase, the project team should have a clear picture of the data's strengths, limitations and potential challenges, enabling informed decisions on how to prepare it for modeling.

3. Data Preparation

The data preparation phase is often the most labor-intensive step of CRISP-DM, sometimes taking up to 70% of the project effort. This step involves cleaning, transforming and structuring the raw data into a form suitable for modeling. Data cleaning focuses on removing duplicate entries, handling missing values through imputation techniques and correcting inconsistencies. Normalization or scaling is applied to bring numerical values into comparable ranges, ensuring no single feature dominates due to larger magnitudes. Categorical data, such as job titles, education levels or employment sectors is transformed into numerical representations through encoding methods such as one-hot encoding or label encoding. Feature engineering is also performed at this stage to enhance predictive power. For example, creating a new feature like "employment duration in current role" or categorizing companies by size or sector can provide additional signals for income prediction. A well-prepared dataset reduces noise, increases accuracy and ensures that the model can learn effectively during training.

4. Modeling

Once the dataset has been prepared, the next step is building predictive or descriptive models. This involves selecting appropriate machine learning algorithms, such as regression models, decision trees, support vector machines or deep learning techniques depending on the problem at hand. Since the objective is to predict numerical salary values, regression-based algorithms are often a natural starting point. The modeling phase often requires trying multiple algorithms to see which performs best on the dataset. Hyperparameter tuning is also a critical component, as different settings like learning rate or regularization strength can significantly affect model accuracy. In many projects, cross-validation is used to test how well the model generalizes across different subsets of the data. The output of this phase is one or more candidate models that capture patterns and relationships between applicant attributes and their estimated income. However, building a technically strong model is not the only priority, it must also remain interpretable and trusted by banking professionals, especially when it influences credit decisions.

5. Evaluation

Evaluation ensures that the chosen model not only performs well statistically but also meets the original business goals defined in the first phase. This involves testing the model on a reserved test dataset that the model has not seen during training. Since salary prediction is a regression task, evaluation metrics such as Root Mean Squared Error (RMSE), Mean Absolute Error (MAE) or R^2 score are typically used. Beyond raw performance, the evaluation phase also examines whether the model aligns with business expectations. For example, whether the salary predictions are realistic, fair across different demographics and useful for loan officers to make sound lending decisions. If the results are unsatisfactory, such as predictions being systematically inaccurate for certain job sectors. The project may loop back to earlier stages like data preparation or modeling for refinement. Only when both business and technical requirements are satisfied can the model move forward for deployment.

6. Deployment

The final phase transitions the project into real-world application. Deployment can range from generating static reports for decision-makers to fully integrating the model into banking systems through APIs during loan application processing. For the salary prediction module, this could mean embedding the model into a loan evaluation platform, where bank staff enter applicant details, such as job, education and experience. The system instantly returns a predicted income value. This output can then be combined with other financial indicators, such as credit scores or debt-to-income ratios, to support loan eligibility decisions. Post-deployment, continuous monitoring is essential to ensure the model remains accurate as job markets, salary benchmarks and applicant profiles evolve. Regular maintenance, retraining and periodic reviews help the system adapt to these changes, while clear documentation, staff training and stakeholder communication ensure the deployed solution remains reliable, transparent and compliant with regulations.

3.1.3 Agentic Artificial Intelligence Cycle



Figure 3.3 Agentic AI Cycle

The Agentic AI Cycle represents the foundational operational framework for autonomous AI systems. It consists of four interconnected and iterative phases: Perceive, Reason, Act and Learn. These phases allow an AI agent to operate in dynamic environments by continuously gathering data, making intelligent decisions, taking meaningful actions and improving over time.

1. Perceive Phase

The Perceive phase is the first and one of the most critical components of the Agentic AI Cycle. It involves the continuous collection of raw data from the agent's environment. In the context of the salary prediction module for loan applications, this includes gathering up-to-date applicant information such as job title, industry, education level, years of experience and geographic location. Data may come from user inputs, bank databases or third-party APIs like employment verification services or economic indicators. The objective is to ensure the AI system has an accurate and real-time understanding of the applicant's context to make informed predictions. The data collected is often unstructured or inconsistent and requires preprocessing steps such as cleaning, filtering or transformation before it can be used for reasoning or prediction. For example, job titles may be standardized, missing fields imputed or categorical features encoded numerically. Without reliable perception, subsequent stages like reasoning or model execution could be compromised. In modern systems, multimodal perception may also be used to integrate structured financial data with external socio-economic trends, further enhancing prediction accuracy.

2. Reason Phase

The Reason phase is where the AI agent processes the perceived data to derive insights and make informed decisions. In the salary prediction module, this phase is responsible for evaluating the input features such as experience level, education, job sector and generating a predicted salary using a trained machine learning model. This phase mimics cognitive functions like pattern recognition and predictive modeling, transforming applicant data into a meaningful income estimate. Various reasoning strategies may be involved, including regression models, ensemble methods or neural networks. The AI also considers historical trends and economic conditions, if available, to contextualize the prediction. Reasoning also supports edge-case detection, for instance, identifying when an input falls outside normal bounds, which may trigger fallback mechanisms or additional checks. This ensures that predictions are not only statistically sound but also reliable and aligned with lending policies.

3. Act Phase

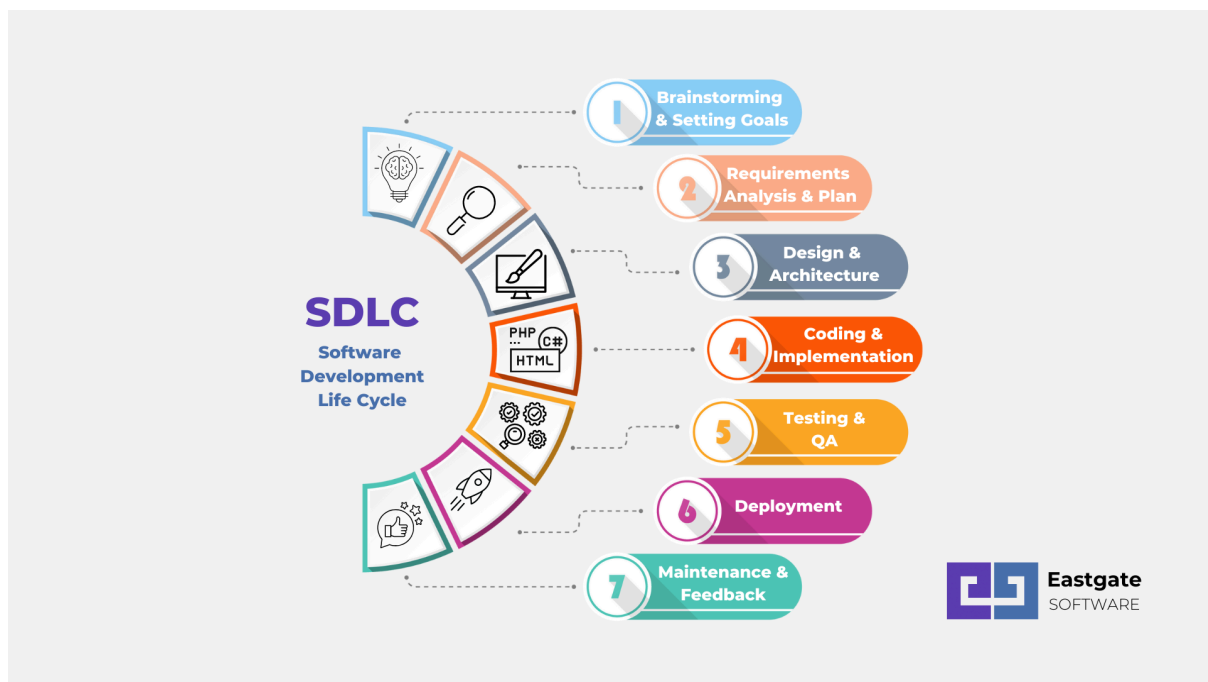
In the Act phase, the AI system carries out the decision derived during the Reason phase by exposing the salary prediction model as an API. This API serves as the main interface through which external systems such as the bank's loan processing platform can interact with the module. When a loan officer or an automated system submits applicant information, for example, job title, education level and experience to the API, it processes the input and returns a real-time predicted salary. This output supports the bank in evaluating the applicant's income level, which in turn helps determine repayment capacity and loan eligibility. The API is designed to be scalable, secure and fast, allowing for high-volume requests with low latency. It enables seamless integration into the bank's existing infrastructure, whether it's a web portal for staff or part of an automated loan evaluation pipeline. Importantly, every API call, along with its input data and returned salary prediction, is logged. These logs form the basis for performance monitoring and provide valuable feedback for the Learn phase, where the model may be retrained based on real-world outcomes. This feedback loop ensures that the salary prediction module evolves and improves over time. In summary, the Act phase in the salary prediction module is focused solely on delivering predictions via a dedicated API, making the model accessible, actionable and responsive to real-time banking needs.

4. Learn Phase

The Learn phase transforms the system from a static tool into a dynamic, self-improving agent. In the salary prediction module, this phase is critical because it involves ongoing model retraining based on real-world outcomes and newly collected data. As more loan applications are processed and actual salary outcomes are collected, the AI system uses this data to update its training dataset. Retraining can be scheduled periodically or occur continuously if the

system supports online or incremental learning. Various machine learning techniques are applied here such as supervised learning using new labeled data or reinforcement learning where the system adjusts based on success metrics like repayment rates. For example, if a predicted high-income applicant defaults, the system adjusts to prevent similar overestimations in the future. This phase also includes model evaluation and tuning, ensuring that new versions maintain fairness, accuracy and compliance. Importantly, as the system learns, the updated model is redeployed via the API, making the entire loop self-sustaining. Thus, the Learn phase completes the Agentic AI Cycle, enabling the module to improve prediction quality, adapt to labor market changes and stay aligned with business goals such as reducing loan defaults or streamlining approval processes.

3.1.4 Software Development Life Cycle



3.4 Process of Software Development Life Cycle

The Software Development Life Cycle (SDLC) is a structured process followed by software development teams to design, develop, test and maintain high-quality software efficiently and predictably. It consists of seven well-defined phases: Brainstorming & Setting Goals, Requirements Analysis & Plan, Design & Architecture, Coding & Implementation, Testing & QA, Deployment and Maintenance & Feedback. Each phase plays a critical role in the software's success, providing clarity, structure and direction from idea to execution.

1. Brainstorming and Setting Goals

This phase defines the purpose, objectives and target users of the system through idea generation and feasibility discussions. By setting SMART goals and assessing risks early, it creates a clear foundation to align stakeholders and avoid future missteps. In this project, the aim was to address inefficiencies and bias in traditional loan approvals by developing an Agentic AI-based credit risk modeling system, with salary prediction as the core module to ensure fairer and more accurate repayment assessments.

2. Requirements Analysis & Plan

In this phase, high-level goals are translated into detailed functional and non-functional requirements, covering features, performance and usability needs. A project plan is then established with priorities, timelines, resources and risk strategies to ensure clarity and shared expectations. For this project, the requirements included accurate salary forecasting, integration into credit scoring workflows and API deployment, supported by non-functional needs such as scalability, transparency and fairness. A plan was drafted to manage risks like data drift and the limitations of synthetic datasets.

3. Design & Architecture

This stage develops the technical blueprint, including system architecture, data flows and user interfaces, ensuring scalability and maintainability. High-level and low-level designs specify the components, logic and tools, translating business needs into a structured technical framework. The design for this project outlined data pipelines, preprocessing steps, regression models for salary prediction and API endpoints, with a modular architecture that supports integration with submodules such as adaptive interest rate modeling and RAG-based decision support.

4. Coding & Implementation

This phase is where the actual development takes place, with code written according to design specifications while following best practices and version control. Agile methods, unit testing and code reviews ensure that the system is functional, reliable and maintainable. For this project, Python was used to implement the salary prediction module, applying regression techniques on synthetic datasets generated with the Faker library, while Scikit-learn and FastAPI supported model training, evaluation and API exposure.

5. Testing and Quality Assurance

The system undergoes rigorous testing to validate correctness, performance, security and usability, using methods such as unit, integration and acceptance tests. Effective QA ensures bugs are detected, requirements are met and the software performs reliably in varied conditions. In this project, model accuracy was validated using RMSE and R^2 metrics, while additional tests examined bias, robustness under data variations and the proper functioning of the API in delivering predictions.

6. Deployment

This phase transitions the system into a live environment where it becomes accessible to users in real workflows. Deployment strategies, such as staging, automation pipelines and rollback options, ensure smooth and reliable releases. For this project, the trained salary prediction model was deployed as a prototype API, demonstrating its ability to provide real-time, data-driven insights for loan approvals in banking operations.

7. Maintenance & Feedback

After deployment, the system is continuously monitored and updated to fix bugs, patch vulnerabilities and enhance features based on user feedback and analytics. This ensures long-term usability, security and adaptability to changing needs. In this project, Agentic AI principles enable continuous retraining of the model to adapt to evolving customer behaviors, market shifts and economic conditions, keeping the system accurate, fair and competitive over time.

3.2 Requirement Gathering Techniques

Requirement gathering is a vital component in developing a system. There are multiple techniques that can be used to gather all the necessary requirements efficiently. In this case, we will be using interviews and questionnaires in order to gather the information and requirements for the Agentic AI Salary Prediction Model.

3.2.1 Interview

An interview is a structured conversation involving either a group of people or a one-on-one interaction. In this case, interviews between the software developers, financial institutions such as commercial banks and fintech platforms as well as their stakeholders will be conducted in order to collect and gather the requirements for the system. Structured questions that are prepared thoroughly will be the key in leading the discussion to gain specific functional requirements and improvements that are needed for the system.

These interviews will involve representatives from banks, loan officers, financial analysts and compliance officers. Through the interviews, stakeholders will be more open to express their views and suggestions, helping the system developers gather requirements that are both business-aligned and user-centered.

One of the key figures that should be prioritized is the bank or fintech loan manager, as they oversee loan approval strategies and policies. Their requirements will ensure that the system aligns with institutional goals and regulatory compliance. Loan officers and credit analysts will also be interviewed, since they are directly involved in assessing applications and will provide practical insights into usability and decision-making support. Compliance officers will provide requirements related to data privacy, auditability and financial regulations. Finally, loan applicants can express their expectations about transparency, fairness and ease of use.

Interview is chosen as a technique due to its effectiveness in facilitating direct communication between the development team and stakeholders. This technique also helps in exploring complex topics, clarifying ambiguities and understanding the regulatory environment of financial institutions. Moreover, interviews provide opportunities to build rapport with stakeholders, fostering an open environment where everyone is comfortable sharing insights and perspectives.

Requirement Gathered

After conducting interviews with financial institution representatives, we gain valuable insights and information about the features that should be implemented in the system. Loan officers highlighted the need for the system to automatically predict an applicant's salary using input features such as employment history, education level and credit background. Compliance officers emphasized the importance of data encryption, secure storage and audit logs to meet regulatory standards. Financial analysts recommended that the system should include explainable AI (XAI) features, allowing both staff and applicants to understand how predictions are made. Additionally, loan applicants expressed that they want the system to be transparent, user-friendly and privacy-protective when their personal information is used.

3.2.2 Questionnaires

Questionnaire is another method that is also significantly important in the requirement gathering process. A questionnaire is a series of questions designed specifically to facilitate the collection of information for research, evaluation or decision-making. For this project, questionnaires will be distributed online using platforms such as Google Forms, which allow efficient collection and analysis

of responses. The questionnaire will include both open-ended and closed-ended questions depending on the type of information being collected.

The respondents will consist of loan applicants (customers), loan officers, financial analysts and compliance teams. For customers, the questionnaire will focus on user experience, trust and fairness in AI-driven predictions. Loan officers and analysts will answer questions related to accuracy thresholds, risk classification and integration needs, while compliance teams will provide input on regulatory and security expectations.

By using questionnaires as one of the requirement gathering techniques, we can easily distribute them to all related groups in a fast and efficient way. This saves time during the requirement phase while still allowing us to gather diverse perspectives. Furthermore, the process of crafting, distributing and analyzing questionnaires is inexpensive, making it one of the most cost-effective techniques compared to others. Online platforms such as Google Forms also provide features for automatically generating data visualizations (charts and graphs), which makes interpretation more straightforward. Importantly, the anonymity of responses encourages stakeholders to provide honest feedback, especially when discussing sensitive topics like privacy and fairness.

Requirement Gathered

After distributing the questionnaires to loan applicants, bank staff and compliance officers, new additional requirements were identified. Loan applicants requested that the system provide clear explanations of how their salary prediction is calculated, so that they can build trust in the process. Loan officers highlighted the need for the system to generate confidence scores and risk categories, assisting them in making more accurate decisions. Compliance teams suggested that the system should include automatic alerts and reporting features to flag potential compliance risks. Meanwhile, financial analysts emphasized the need for periodic retraining of the AI model to maintain prediction accuracy as labor market and economic conditions change.

3.3 Functional and Non Functional Requirement

3.3.1 Functional Requirements

1. Data Input and Preprocessing

- The system shall provide a user-friendly interface for applicants to input personal and financial information, including income, employment details, expenses, credit score and other relevant data.

- The system shall validate, clean and preprocess user input to ensure data integrity before feeding it to the prediction module.
- The system shall store each new record in the database, tagged for future use in model evaluation and retraining.

2. Salary Prediction Using Agentic AI Model

- Upon receiving user input, the system shall immediately run the data through an agentic AI-based predictive model to estimate the applicant's future salary.
- The system shall present the predicted salary along with a confidence score and eligibility status (Eligible, Conditional, Not Eligible).
- The prediction output shall be integrated into the loan decision workflow and available to bank officers via the dashboard.

3. Continuous Learning and Model Retraining

- The system shall automatically log new input records and their final loan outcomes (approved, rejected, defaulted) for model improvement.
- The agentic AI model shall periodically retrain using newly collected and verified data to improve future prediction accuracy.
- Retraining can occur on a scheduled basis or be triggered by a defined threshold of new data points.
- The system shall evaluate retrained models using validation metrics before deployment.

4. Loan Decision Integration

- The predicted salary shall be used to calculate repayment capacity and determine loan eligibility.
- The system shall integrate salary predictions and risk scores into the loan approval process, triggering automatic or manual decision paths.

5. Risk Assessment and Scoring

- The system shall compute a risk score for each applicant based on predicted salary, debt-to-income ratio, financial behavior and credit history.
 - Risk scores shall be categorized as low, medium or high and displayed on the dashboard with explanations of key risk factors.
-

6. Reporting Dashboard for Bank Officers

- The system shall offer a real-time dashboard that displays applicant details, predicted salaries, eligibility status and risk assessments.
- Officers shall be able to filter applications by risk level, prediction confidence or other custom parameters.
- Insights and recommendations shall be generated to support manual reviews and final loan decisions.

7. AI Chatbot Integration

- An integrated AI chatbot shall guide users through the application process, answer eligibility-related questions and offer tips for improving input quality.
- For bank staff, the chatbot shall function as a smart assistant, capable of fetching prediction results, applicant summaries or system recommendations on request.

3.3.2 Non - Functional Requirements**1. Product**

The Salary Prediction Module is designed with high usability in mind to support users in determining their loan eligibility. The user interface (UI) is simple and intuitive, allowing users to enter personal information and financial history with ease. It incorporates user guidance and familiar visual elements, ensuring a minimal learning curve for individuals of varying technical backgrounds. The system is optimized to process this input data and deliver accurate salary predictions within a few seconds under normal operating conditions. It is capable of handling large datasets without performance degradation, ensuring smooth operation even during peak usage.

Accuracy and reliability are core to the system's function. The module uses predictive algorithms to estimate the applicant's salary based on their input, which then informs whether they are eligible for a loan. To minimize service interruptions, the system is built to maintain an uptime of at least 99.95%. The prediction results must be consistent and reliable, as they directly impact critical financial decisions. Additionally, efficient and scalable data storage is ensured so that the system can grow with user demand while maintaining quick access and low resource usage.

2. External

The module handles sensitive personal and financial data and as such, user privacy is a top priority. Only essential information such as user ID and final prediction summary may be accessed by authorized staff. All personal and financial information provided, including income, expenses, employment details, credit score and contact information, must be securely stored and used strictly for the purpose of salary prediction and loan eligibility analysis. Unauthorized access or misuse of this data could lead to identity theft, financial fraud or reputational harm, so strict privacy controls and data protection mechanisms are enforced.

System safety is also critical. Strong security measures are implemented to prevent unauthorized access and safeguard sensitive data. Authentication mechanisms ensure that only verified users can log in and interact with the system. Role-based access control restricts different types of users, such as the staff, managers and the officers to only the information and actions relevant to their roles. These protections are especially important during data entry and when handling prediction results, as breaches at these points could compromise user trust and system integrity. Overall, the system is built to support secure, trustworthy and accurate predictions that enable informed decisions about loan approvals.

3.4 Description of Dataset

The dataset utilized in this salary prediction module is a hybrid compilation of real-world and synthetically generated data. The core dataset was sourced from Kaggle, a widely recognized platform for open datasets used in data science and machine learning projects. This foundational dataset includes structured tabular information covering individual-level salary details and related employment features such as education, job title and industry.

To enhance the richness and contextual relevance of the dataset, additional fields were generated using the Faker Python library. The Faker-generated attributes simulate realistic employment history and financial background information that are typically not available in public datasets due to privacy concerns. These synthetic data points include credit score, years at previous jobs, number of dependents and prior annual income, all formatted to be statistically coherent with the original dataset.

By combining real and synthetic data, the resulting dataset is both comprehensive and privacy-compliant, providing a broader set of features useful for predictive modeling, particularly in supervised machine learning tasks such as regression-based salary prediction. The final dataset is stored in CSV format and adopts a relational tabular structure, where each row represents an

individual record and each column denotes a distinct feature. The dataset has been preprocessed for consistency, including handling of missing values, normalization of categorical variables and encoding of dates or ordinal variables where applicable. It is readily loadable using the pandas library in Python, making it easy to integrate into machine learning pipelines.

This structured and extensible dataset supports various modeling scenarios including salary estimation, employment risk assessment and financial profiling, offering strong potential for both academic and industry research use cases.

Source : <https://www.kaggle.com/datasets/mrsimple07/salary-prediction-data>

3.4.1 Data Structure and Data Dictionary

Table 3.1 Data Structure and Data Dictionary of Dataset

Data Attribute	Data Type	Description	Source
Education	Categorical (String)	Highest level of education attained	Kaggle
Experience	Numerical (Integer)	Total years of professional experience	Kaggle
Location	Categorical (String)	City or region of the applicant	Kaggle
Job Title	Categorical (String)	Current Job Position	Kaggle
Age	Numerical (Integer)	Age of the applicant	Kaggle
Gender	Categorical (String)	Gender of the applicant	Kaggle
Salary	Numerical (Double)	Current Salary of applicant	Kaggle
Industry	Categorical (String)	Employment industry	Generate using Faker Library
Employment Type	Categorical (String)	Type of employment	Generate using Faker Library
Prior Annual Income	Numerical (Double)	Annual income from previous employment	Generate using Faker Library
Years at Last Job	Numerical (Integer)	Years of previous job held	Generate using Faker Library
Financial Dependents	Numerical (Integer)	Number of individuals financially dependent on the applicant	Generate using Faker Library

Marital Status	Categorical (String)	Marital Status of Applicant	Generate using Faker Library
----------------	----------------------	-----------------------------	------------------------------

3.5 Chapter Summary and Evaluation

The chapter provides a comprehensive explanation of the methodology and system design for the salary prediction module in loan applications, framed within multiple structured approaches such as the system flowchart, CRISP-DM methodology, Agentic AI Cycle and SDLC framework. The flowchart describes the complete workflow starting from data collection, preprocessing and feature extraction to model training, evaluation and hyperparameter tuning, ultimately leading to deployment for real-world loan assessments. This is complemented by the CRISP-DM methodology, which ensures alignment between business goals and technical processes, highlighting phases such as business understanding, data preparation, modeling and evaluation. The Agentic AI Cycle introduces adaptability and autonomy through its Perceive, Reason, Act and Learn stages, ensuring continuous improvement of the predictive model. Finally, the SDLC phases from brainstorming and design to testing, deployment and maintenance. It demonstrates how the system is built to be scalable, secure and user-friendly. Together, these methodologies provide a clear picture of the system's robustness, ensuring accuracy, reliability and compliance with data security standards.

In evaluation, the chapter effectively integrates traditional data science methodologies with advanced AI-driven frameworks, demonstrating both technical rigor and practical relevance. The use of CRISP-DM ensures systematic handling of data mining tasks, while the Agentic AI Cycle emphasizes adaptability by enabling continuous retraining and performance improvement. The inclusion of functional and non-functional requirements strengthens the system's design by balancing predictive accuracy with usability, security and scalability. Moreover, the combination of real-world and synthetic data broadens the dataset's richness while maintaining privacy compliance, which is crucial in financial applications. However, while the methodologies are well-aligned, the reliance on synthetic data may introduce bias if not carefully validated and the complexity of maintaining continuous retraining could pose challenges in real-world banking environments. Overall, the chapter presents a solid framework that merges structured methodologies with adaptive AI capabilities, making it both academically rigorous and practically applicable for enhancing loan decision-making processes.

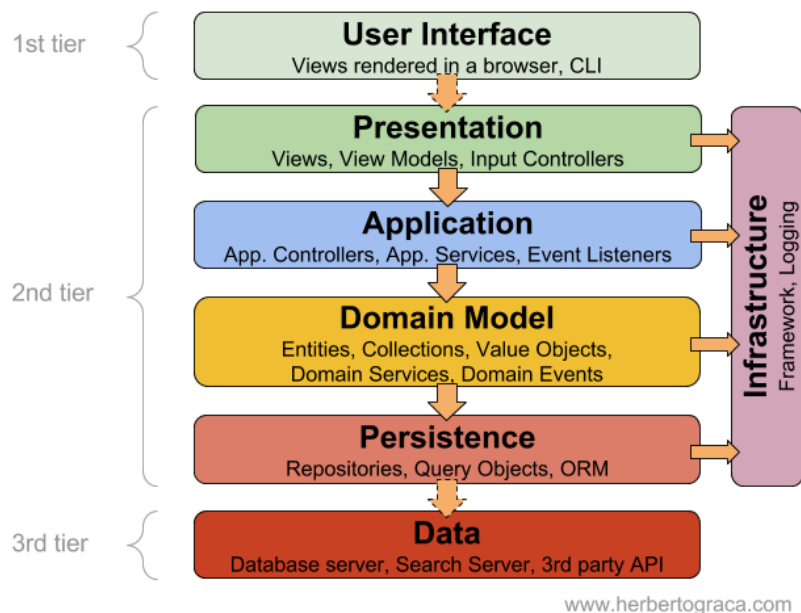
Chapter 4

System Design

4 System Design

This chapter provides an overview of the system architecture, algorithmic design and user interface design for the Salary Prediction Model. It explains how the system is structured to handle applicant information, process it through different layers and generate meaningful outputs such as salary predictions and risk assessments. The design focuses on creating a framework that is reliable, scalable and transparent, while ensuring ease of use for both applicants and financial staff. It also outlines the main methods and techniques applied in the project, showing how they work together to support accurate and understandable decision-making.

4.1 System Architecture Design



4.1 System Architecture Design

4.1.1 User Interface - 1st Tier

The user interface serves as the primary access point for both loan applicants and financial institution staff. Loan applicants interact with the system through a web or mobile application where they can input personal and financial details such as income, expenses, credit score and employment history. Loan officers, on the other hand, access a real-time dashboard that displays applicant details, predicted salaries, risk assessments and SHAP-based explanations, which highlight the key factors influencing each prediction. The chatbot is also embedded within this interface, enabling applicants to ask questions about their eligibility or predicted salary, while loan officers can query the system for quick summaries of applicant risk and loan statuses. By ensuring an intuitive and user-friendly experience, the user interface promotes transparency, accessibility and engagement across all user groups.

4.1.2 Presentation Layer - 2nd tier

The presentation layer acts as the mediator between the user interface and the backend services, ensuring that data is validated, processed and displayed in a structured manner. When applicants submit their details, the presentation layer validates inputs such as employment details or credit scores, then prepares them for secure transmission via the API. It also formats and visualizes outputs, including prediction results, confidence levels, risk categories and SHAP explanatory graphs that make the AI's decision process understandable. For the chatbot, this layer renders natural language responses, ensuring that interactions feel seamless and conversational. By managing both input validation and output presentation, this layer guarantees that users receive accurate and comprehensible information.

4.1.3 Application Layer - 2nd Tier

The application layer orchestrates the entire workflow of the salary prediction module, ensuring smooth communication between user inputs, domain logic and external services. It handles API calls to the agentic AI salary prediction service, sending validated applicant data for real-time salary estimation. This layer also invokes the explainability service powered by SHAP, generating feature contribution explanations to accompany predictions. In addition, it integrates with the chatbot service, allowing dynamic question-and-answer functionality related to applicant eligibility, prediction outcomes and compliance status. The application layer then routes all results predictions, explanations and chatbot responses back to the presentation layer, ensuring that users receive relevant and timely insights.

4.1.4 Domain Model Layer - 2nd Tier

The domain model layer contains the core business logic and intelligence of the salary prediction system. It defines the main entities such as applicants, loan applications, salary predictions and risk scores and it governs the rules that determine eligibility and repayment capacity. The agentic AI salary prediction model resides here, processing applicant features and generating predictions that feed into risk categorization services. SHAP-based explainability logic is also embedded in this layer, ensuring that every prediction is accompanied by transparent reasoning. Domain events, such as "Salary Predicted" or "High-Risk Applicant Flagged" are triggered here to support decision workflows in banking systems. This makes the domain layer the heart of the system, where both predictive intelligence and business-driven logic converge.

4.1.5 Persistence Layer - 2nd Tier

The persistence layer manages data storage and retrieval, ensuring that applicant information, prediction history, SHAP values and chatbot conversations are securely stored for future use. It employs repositories and query objects to fetch applicant details, store prediction outcomes and update compliance logs in real time. This layer also supports the continuous learning mechanism of the agentic AI model by storing new applicant data and loan outcomes, which can later be used for retraining. By maintaining structured, reliable and scalable data access, the persistence layer guarantees the integrity and availability of records critical to both predictions and decision-making.

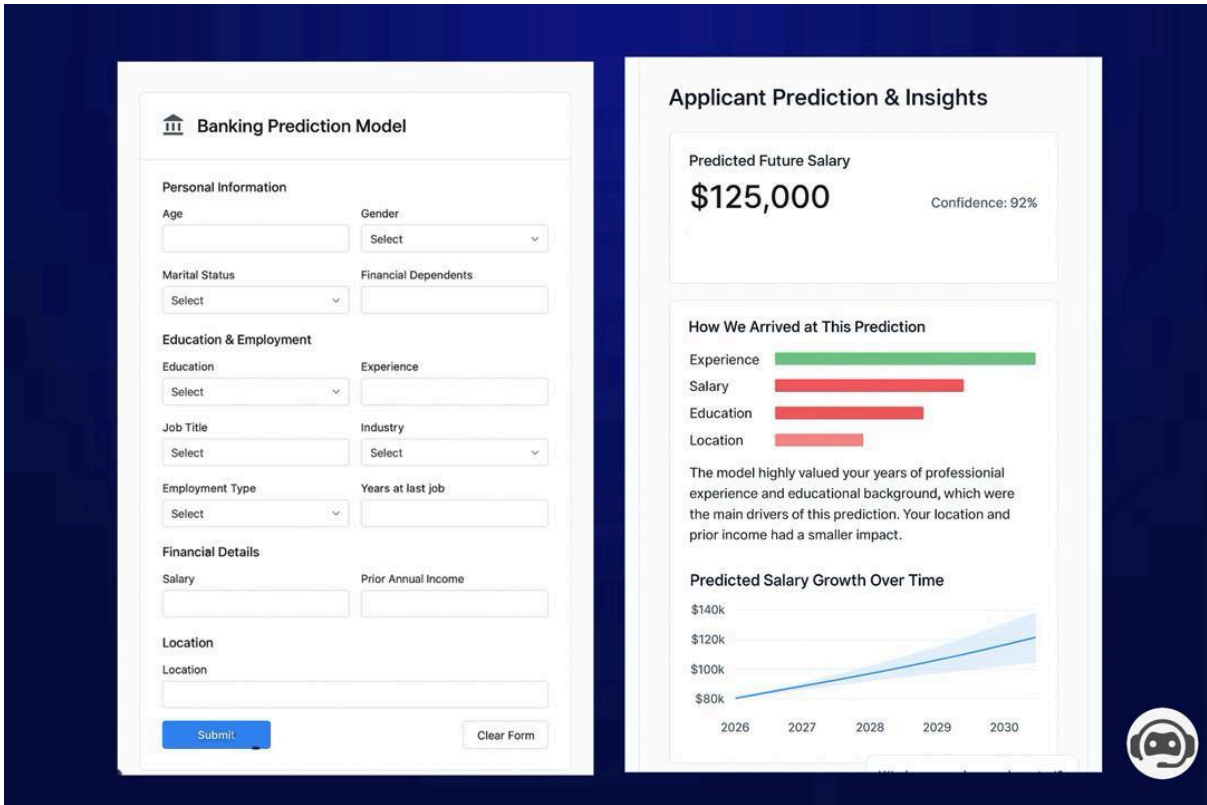
4.1.6 Data Layer - 3rd Layer

The data layer provides the foundational storage and external data connections required for the system to function effectively. A centralized database houses applicant records, prediction outputs, SHAP explanation data and compliance logs, ensuring that sensitive financial information is securely maintained. Additionally, the system can integrate with external APIs to enrich predictions for example, by pulling labor market benchmarks, economic indicators or credit bureau data to provide more accurate salary estimations. Search servers may also be employed to quickly retrieve historical applicant records or compliance audit trails. By combining internal storage with external data sources, this layer ensures that the AI model is both well-informed and grounded in real-world financial contexts.

4.1.7 Infrastructure

The infrastructure layer underpins the entire system by providing essential services such as frameworks, logging, monitoring and security. An API gateway manages secure communication between the presentation, application and domain layers, ensuring that the prediction and explainability services are modular and accessible. Logging mechanisms capture key events, such as prediction requests, chatbot interactions and compliance alerts, while monitoring tools track system health and performance. Security is enforced through encryption, authentication and role-based access control, ensuring that sensitive applicant and prediction data are only accessible to authorized users. By enabling scalability, compliance and trust, the infrastructure layer ensures that the salary prediction module operates reliably within a financial environment.

4.2 User Interface (UI) Design



4.2 User Interface Design of the Salary Prediction Module in Credit Risk Modeling System

The figure 4.2 demonstrates the User Interface Design of Salary Prediction Module in Credit Risk Modeling System. It provides a structured form where users can enter their personal information like age, gender, marital status, dependents, location, education and employment details as well as financial details. After clicking on the action button ‘Submit’, the system will process the inputs and produce insights for decision making. The system displays the predicted future salary along with a confidence score (92%). Below, it provides explainability insights with a bar chart showing which factors contributed most to the prediction. A short explanation clarifies that experience and current salary greatly impact the decision from the figure above. Finally, a line chart projects salary growth over time, giving the user a forward-looking view of income potential.

4.3 Algorithm and Method Used

Table 4.1 Algorithms and Method Used in Project

Algorithms/ Method	Purpose	Application in Project	Usage in Project
Random Forest	To make accurate predictions	Used to predict applicant salary	Trained on historical salary

Regressor	by combining the results of multiple decision trees and reducing overfitting	based on various features like education level, job role, experience, etc.	data; outputs a reliable salary prediction for each loan applicant via the FastAPI module
XGBoost (Gradient Boosting)	To enhance prediction accuracy by building trees sequentially and correcting previous errors	Applied as an advanced alternative to Random Forest for structured data like applicant profiles	Trained and tuned to deliver high-performance salary predictions with better generalization on unseen applicant data
FastAPI (RESTful API)	To expose the trained model as an HTTP endpoint for real-time predictions	Integrates the salary prediction model into the bank's loan processing system	Receives applicant data via POST request and returns a real-time salary prediction in JSON format
SHAP (SHapley Additive exPlanations)	To interpret model predictions and explain the contribution of each feature	Provides transparency to salary predictions by showing which applicant attributes (e.g., education, experience) influenced the outcome most	Integrated into the model evaluation pipeline to generate explanations for both developers and banking officers

4.4 Chapter Summary and Evaluation

In summary, the chapter provides a structured overview of the system's architecture, highlighting the responsibilities of each tier from the intuitive user interface to the data and infrastructure layers that ensure security, scalability and compliance. The algorithm design section demonstrates the thoughtful selection of predictive and interpretability tools, balancing accuracy with transparency through SHAP explanations and sound preprocessing methods. Overall, the design achieves a strong alignment between technical rigor and practical banking requirements, ensuring that predictions are not only accurate but also explainable and trustworthy. This balance positions the model as a reliable decision-support tool that can enhance both applicant experience and institutional efficiency.

References

Abdou, H. A., & Pointon, J. (2011). CREDIT SCORING, STATISTICAL TECHNIQUES AND EVALUATION CRITERIA: A REVIEW OF THE LITERATURE. *Intelligent Systems in Accounting, Finance and Management*, 18(2-3), 59–88. <https://doi.org/10.1002/isaf.325>

Abdou, H. A., Tsafack, M. D. D., Ntim, C. G., & Baker, R. D. (2016). Predicting creditworthiness in retail banking with limited scoring data. *Knowledge-Based Systems*, 103, 89–103. <https://doi.org/10.1016/j.knosys.2016.03.023>

Agentic AI in financial service. (2025). Moodys.com. <https://www.moodys.com/web/en/us/creditview/blog/agentic-ai-in-financial-services.html>

Aldasoro, I., Fender, I., Hardy, B., & Tarashev, N. (2020, May 7). Effects of Covid-19 on the banking sector: the market's assessment. Ideas.repec.org. <https://ideas.repec.org/p/bis/bisblt/12.html>

An Introduction to MultiAgent Systems. (2025). Google Books. <https://books.google.com.my/books?hl=en&lr=&id=X3ZQ7yeDn2IC&oi=fnd&pg=PR13&dq=Wooldridge>

Bharath Somu. (2024). The Future of Banking IT Services: Convergence of Intelligent Infrastructure and Agentic AI Models. *Global Research Development (GRD)* ISSN: 2455-5703, 9(12). <https://grdjournals.com/index.php/grdjournal/article/view/133>

Brevoort, K., Holmes, D. B., & Wolter, S. B. (2016). An economic analysis of the impact of the Fair Credit Reporting Act on consumers (Loyola University Chicago Law Review, Paper No. 1667). *LAW eCommons*. <https://lawecommons.luc.edu/cgi/viewcontent.cgi?article=1667&context=lclr>

Bussmann, N., Giudici, P., Marinelli, D., & Papenbrock, J. (2020). Explainable AI in Fintech Risk Management. *Frontiers in Artificial Intelligence*, 3(26). <https://doi.org/10.3389/frai.2020.00026>

Carter, A. (n.d.). JOURNAL OF COMPARATIVE RESEARCH IN ANTHROPOLOGY AND SOCIOLOGY. https://www.alexcarterla.com/uploads/1/1/7/6/117642977/oneil_-_eubanks.pdf

Casolaro, A., Capone, V., Iannuzzo, G., & Camastra, F. (2023). Deep Learning for Time Series Forecasting: Advances and Open Problems. Information, 14(11), 598. <https://doi.org/10.3390/info14110598>

Chen, T., & Guestrin, C. (2016). XGBoost: a Scalable Tree Boosting System. Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining - KDD '16, 1(1), 785–794. <https://doi.org/10.1145/2939672.2939785>

Consumer Credit Models. (2025). Google Books. <https://books.google.com.my/books?hl=en&lr=&id=zH3OHkVQZFwC&oi=fnd&pg=PR11&dq=Thomas>

Fairness and Machine Learning. (2023). Google Books. <https://books.google.com.my/books?hl=en&lr=&id=HuGwEAAAQBAJ&oi=fnd&pg=PR9&dq=Barocas>

Gomez-Cravioto, D. A., Diaz-Ramos, R. E., Hernandez-Gress, N., Preciado, J. L., & Ceballos, H. G. (2022). Supervised machine learning predictive analytics for alumni income. Journal of Big Data, 9(1). <https://doi.org/10.1186/s40537-022-00559-6>

Goodell, J. W. (2020). COVID-19 and finance: Agendas for future research. Finance Research Letters, 35(101512), 101512. <https://doi.org/10.1016/j.frl.2020.101512>

Hand, D. J., & Henley, W. E. (1997). Statistical Classification Methods in Consumer Credit Scoring: a Review. Journal of the Royal Statistical Society: Series a (Statistics in Society), 160(3), 523–541. <https://doi.org/10.1111/j.1467-985x.1997.00078.x>

Khandani, A. E., Kim, A. J., & Lo, A. W. (2010). Consumer credit-risk models via machine-learning algorithms. *Journal of Banking & Finance*, 34(11), 2767–2787. <https://doi.org/10.1016/j.jbankfin.2010.06.001>

Jagtiani, J., & Lemieux, C. (2019). The Roles of Alternative Data and Machine Learning in Fintech lending: Evidence from the LendingClub Consumer Platform. *Financial Management*, 48(4), 1009–1029. <https://doi.org/10.1111/fima.12295>

Joshi, M., Savita Bhosale, & Vyawahare, V. A. (2022). Using fractional derivative in learning algorithm for artificial neural network: Application for salary prediction. <https://doi.org/10.1109/ibssc56953.2022.10037558>

Lessmann, S., Baesens, B., Seow, H.-V., & Thomas, L. C. (2015). Benchmarking state-of-the-art classification algorithms for credit scoring: An update of research. *European Journal of Operational Research*, 247(1), 124–136.

Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küttler, H., Lewis, M., Yih, W., Rocktäschel, T., Riedel, S., & Kiela, D. (2020). Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. *Neural Information Processing Systems*; Curran Associates, Inc. <https://proceedings.neurips.cc/paper/2020/hash/6b493230205f780e1bc26945df7481e5-Abstract.html>

Mincer, J. A. (1974). The Human Capital Earnings Function. *NBER Chapters*, 83–96. <https://ideas.repec.org/h/nbr/nberch/1767.html>

Mishra, P., Srivastava, S., Gupta, P., Anand, A., & Gupta, S. C. (2021, December 1). A Comparative Study of Machine Learning Algorithms for Salary Estimation. *IEEE Xplore*. <https://doi.org/10.1109/ICAC3N53548.2021.9725775>

Park, J. S., O'Brien, J. C., Cai, C. J., Morris, M. R., Liang, P., & Bernstein, M. S. (2023). Generative Agents: Interactive Simulacra of Human Behavior. *ArXiv:2304.03442 [Cs]*. <https://arxiv.org/abs/2304.03442>

Rudin, C. (2019). Stop Explaining Black Box Machine Learning Models for High Stakes Decisions and Use Interpretable Models Instead. ArXiv:1811.10154 [Cs, Stat], 1. <https://arxiv.org/abs/1811.10154>

Samuel Iorhemen Ayua, Yusuf Musa Malgwi, & Afrifa, J. (2023). Salary Prediction Model for Non-Academic Staff Using Polynomial Regression Technique. Artificial Intelligence and Applications. <https://doi.org/10.47852/bonviewaia3202795>

Scheffler, P. (2025, September 5). Agentic AI in Banking: Strategy Guide for C-Level Leaders. Neontri. <https://neontri.com/blog/agentic-ai-banking/>

Wachter, S., Mittelstadt, B., & Russell, C. (2017). Counterfactual Explanations without Opening the Black Box: Automated Decisions and the GDPR. Harvard Journal of Law & Technology (Harvard JOLT),31,841. <https://heinonline.org/HOL/LandingPage?handle=hein.journals/hjlt31&div=29&id=&page=>